1.2 DNA

Thursday, 29 January 2026 12:24 PM

1.2 Observe and model the arrangement of genetic information in an organism to define and compare the terms DNA, gene, chromosome and genome

https://www.clickview.net/interactive/90331111/l

Activity:

Fill in the following glossary terms using your textbook key terms (chapter 5.1)

DNA Gene Chromosome Genome

Summarise the following information in 20 words or less

Note-taking tip

Summarising

Summarising helps you capture key points in your own words. The Cornell note-taking method uses summaries to synthesise and review the main ideas in your notes.

If you can't decide what's most important, try looking for key terms and explaining how they connect to the text.

DNA is a complex molecule containing the genetic information to build nearly all living things. DNA stands for deoxyribonucleic acid. The DNA of an organism provides the blueprint which will determine how it will develop and function. DNA determines the order amino acids will join together to form proteins. The proteins produced can help make cells, which grouped together form tissues, then organs and entire living organisms.

DNA → blueprint

3 base pairs → code for 1 amino acids.

Word count:

amino acid

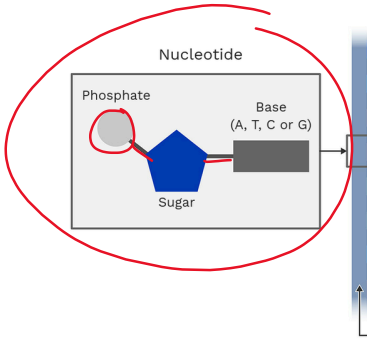


Identify the different parts of a DNA molecule by labelling the diagram and completing the table



Class notes:

- DNA has a doc
- The backbone c of repeating s
- Adenine always bon
- Cytosine always bon
- A, T, C and G are
- One base + one sug



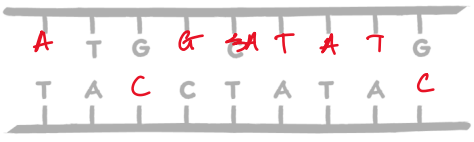
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Structure	Description
double helix	twisted structure of DNA molecule
backbone	sugar phosphate
complimentary base pairs	two bases that combine together
gene	small section of DNA, codes for a trait/
base	nucleic acid A, T, C, G

yy

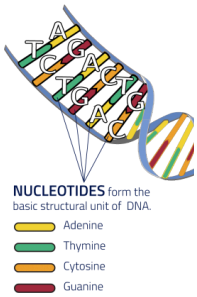
Here is an incomplete segment of DNA. **Demonstrate** your understanding of complementary base pairing by completing the segment and fixing any mistakes. Annotate around the outside to **explain** any rules you followed.



Draw a diagram to represent the arrangement of genetic information in an organism. Your diagram should include the following: a nucleus containing the genome, a chromosome, double helix DNA, a gene, and nucleotides. Use your textbook or the following website for guidance: https://mea.digital/yr3

Links and simulations:

1. https://my.clevelandclinic.org/health/bo
2. https://biomanbio.com/HTML5Gamesa page.html
3. https://www.labxchange.org/library/iten fullscreen=true
4. https://gizmos.explorelearning.com/finc

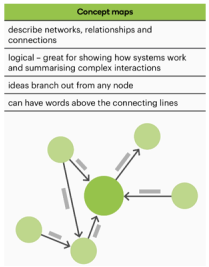


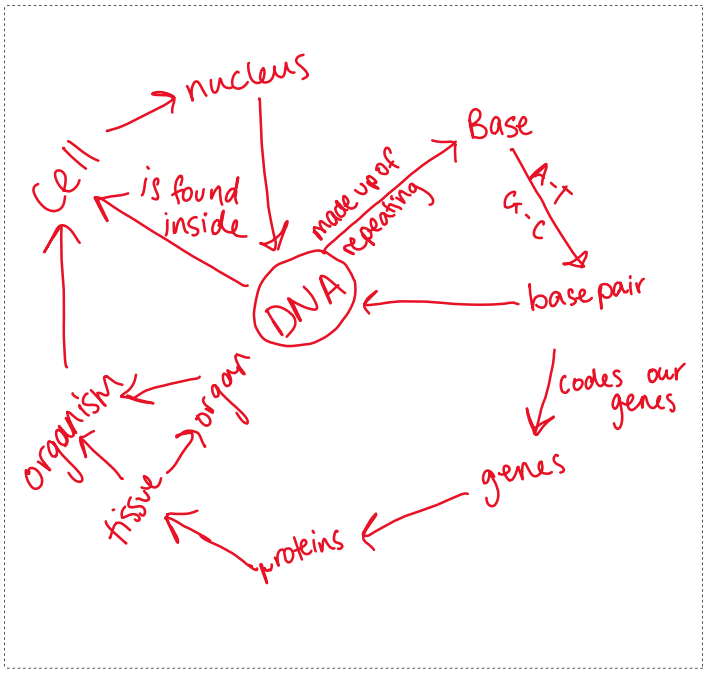
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Create a concept map that links the terms in the box.

Use arrows with words above them to explain how different terms relate to one another. Include their function and how they contribute to the other levels of organisation.

base | base pair | gene | organ | cell
nucleus | protein | organism | tissue | DNA

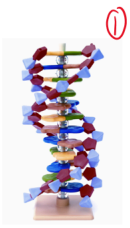


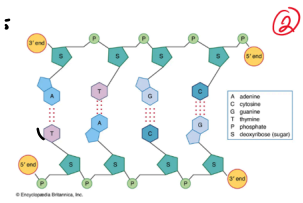


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DNA Models

a) Annotate each image below to indicate what is being represented within each component of the DNA models below.





b) Discuss for each of the three images above, the inclusions and exclusions of their representation of DNA.

DNA models

Using models in s
that we could not
can support a goo

Image	Inclusions	Exclusions
1	double helix shape	
2	legend - labelling provides more detail about DNA	does not show arrangement shape → simplified

limitations in their
certain details. Th
models in science

Allows use to sin and phenomena
Can be used to r hypotheses
Enables us to ex be difficult or imp life
Provides a visua system being stu and communicat

Homework activity – Building a DNA model

Plan and construct a model of a DNA molecule that contains at least 4 base pairs.

Your model must represent the following features:

- Sugar-phosphate backbone
- All four bases (A, T, G, C)
- Complementary base pairing
- Hydrogen bonds between base pairs

My plan:

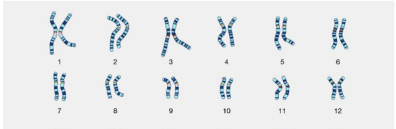
My model:

Print out and stick in a labelled photo of your model

Or upload a photo here.

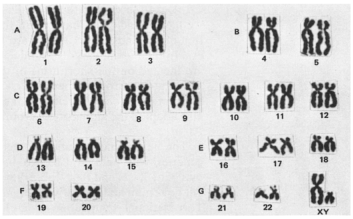
Karyotypes

This is a diagram of a human 'karyotype'. A karyotype is a visual map of the chromosomes in a cell. For humans, a karyotype has 46 chromosomes (2n=46). There are 23 pairs, each numbered according to size. Pairs 1-22 are called



autosomes, while the 23rd pair are the sex chromosomes (X & Y chromosomes). Having XX makes someone a female, while having XY makes someone male.





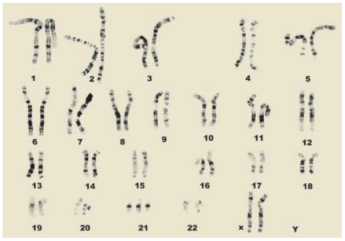
This is also a human karyotype, but each chromosome has been replicated. This happens prior to cell division.

Is this karyotype of a male or female?

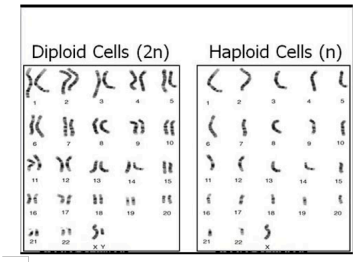
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How do you know?

.....



Can you identify the disorder in this karyotype? Circle it.

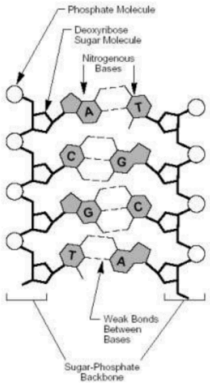


Somatic cells are diploid (2n=46), which means all chromosomes are found in pairs.

Gametes (sperm and eggs) are n = 23, which means they have 23 unpaired chromosomes. When two gametes fuse during fertilisation, the resulting cell has 2n =46.

DNA Fact Sheet

- **DNA** stands for the chemical deoxyribonucleic acid.
- DNA is the chemical which stores genetic information in all living cells. **DNA** from all organisms is made up of the same chemical and physical components.
- **Double helix structure**: DNA can be pictured as an extremely long ladder that has been twisted into the double helix shape or spiral staircase.
- The **sides** of the ladder or double helix are a linked chain of alternating **sugar and phosphate** molecules.
- The **rungs** of the ladder are made of a pair of molecules called **bases** that are weakly attracted to each other.
- There are 4 different bases- **adenine, thymine, cytosine, and guanine** (A,C,T, G for short).



- The 4 DNA bases can pair with each other, but only in certain combinations. A pairs with T. C pairs with G.
- A:T and C:G are said to be **complementary base pairs**
- The 2 DNA strands in a helix are **complementary strands**
- The sequence of the bases determines the genetic information. The **DNA sequence** is the particular side-by-side arrangement of bases along the DNA strand (e.g., ATTCCGGA). This order spells out the exact instructions required to create a particular organism with its own unique traits.